Homework Section 14.2

1. Given $\lim_{(x,y)\to(2,3)} f(x,y) = 8$, under what condition does $f(2,3) = 8$?

2. Find the limit if it exists. If the limit does not exist, then find two paths in the *xy*-plane that produce different limits in order to verify that it doesn't exist.

a) $\lim_{(x,y)\to(-2,4)} 2x^2y - 5x^3y^2 + 7$

b) $\lim_{(x,y)\to(0,0)} \dfrac{3+2y}{\cos(3x-2y)}$

c) $\lim_{(x,y)\to(0,0)} \dfrac{3x^2}{2x^2+y^2}$

d) $\lim_{(x,y)\to(0,0)} \dfrac{x^4-y^4}{x^2+y^2}$

e) $\lim_{(x,y)\to(0,0)} \dfrac{3x^2y}{x^4+y^2}$

f) $\lim_{(x,y,z)\to(-2,4,2)} 2x^2y - 5x^3y^2z - 7z$

3. Let $g(t) = t^2 + \sqrt{t}$ and $f(x,y) = 2x + 3y - 6$. Find $h = g(f(x,y))$ and the set on which h is continuous.

4. For which sets of points are the following functions continuous?

a) $f(x,y) = \dfrac{x}{\sqrt{x+y-2}}$

b) $f(x,y) = \sin^{-1}(x^2+y^2)$

c) $g(x,y) = \ln(x^2+y^2-4)$

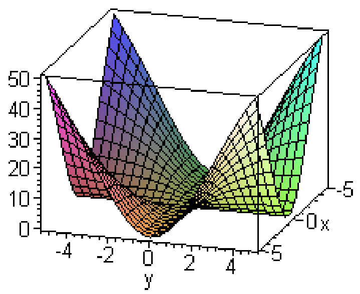
d) $f(x,y) = \begin{cases} \dfrac{xy}{x^2+xy+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

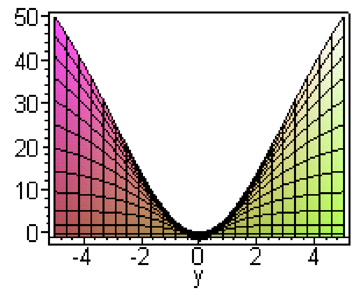
5. Consider the function $f(x,y) = \dfrac{xy}{\sqrt{x^2+y^2}}$. Fill in the below table with function values accurate to the nearest thousandth and then make a conjecture about the value of $\lim_{(x,y)\to(0,0)} \dfrac{xy}{\sqrt{x^2+y^2}}$ **[See the Excel video]**

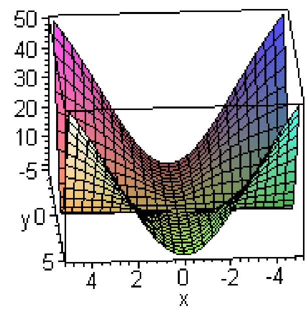
y / x	**-0.2**	**-0.1**	**-.05**	**0**	**0.05**	**0.1**	**0.2**
-0.2							
-0.1							
-0.05							
0							
0.05							
0.1							
0.2							

6. Below are three different views of the graph of $f(x,y)=\dfrac{4x^2y^2}{x^2+y^2}$.

a) Use these graphs to make a conjecture about the value of $\lim\limits_{(x,y)\to(0,0)} \dfrac{4x^2y^2}{x^2+y^2}$. Then explain how you got your answer.







b) Now consider the function $g(x,y)=\begin{cases} \dfrac{4x^2y^2}{x^2+y^2}, & (x,y)\neq(0,0) \\ 0, & (x,y)=(0,0) \end{cases}$. Based on your answer from part (a), is $g(x,y)$ continuous at (0,0)? Explain your answer using the definition of a continuous function at a point.

7. The following is the official definition of a **linear transformation**:

A function T mapping R^n to R^m (n =1,2, or 3, m = 1,2, or 3) is called a **linear transformation** if it satisfies

i. $T(\mathbf{u}+\mathbf{v})=T(\mathbf{u})+T(\mathbf{v})$ for all vectors **u**, **v** in R^n;

ii. $T(c\mathbf{u})=cT(\mathbf{u})$ for all vectors **u** in R^n and scalars c.

In this class a linear transformation T on a column vector **u** with n components is synonymous with left-hand multiplication of **u** (i.e. A**u**) by an $m\times n$ matrix A with constant entries. For example, multiplication by $A=\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ reflects points in R^2 across the y-axis.

Exercise: a) Graph $\mathbf{r}(t)=\langle t,t^3\rangle$ for $t\geq 0$.

b) Note that $\mathbf{r}(t) = \langle t, t^3 \rangle$ can be written as a column vector $\mathbf{r}(t) = \begin{bmatrix} t \\ t^3 \end{bmatrix}$. Find the following matrix product and call it $\mathbf{s}(t)$:

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} t \\ t^3 \end{bmatrix}$$

c) Graph $\mathbf{s}(t)$, $t \geq 0$, on the same plane as $\mathbf{r}(t)$. Note that $\mathbf{s}(t)$ is the reflection of $\mathbf{r}(t)$ across the y-axis.